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RESEARCH ARTICLE

Unveiling the Hidden Interface: Pre-SEI Governs Li Morphology in Anode-Free Li Metal Batteries

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ABSTRACT

Anode-free lithium metal batteries (AFLMBs) are emerging as a promising next-generation energy storage system due to their high energy density and simplified manufacturing process. However, the absence of excess lithium makes them highly vulnerable to dendritic growth and irreversible lithium loss, leading to rapid capacity decay. Despite extensive efforts in current collector modification and electrolyte optimization, the interfacial chemistry between the electrolyte and current collector remains poorly understood. Here, we reveal that a pre-solid electrolyte interphase (pre-SEI) can spontaneously form on the Cu surface during the resting stage, governed by the electrolyte composition. Aggregated anion–cation complexes (AGGs) with low lowest unoccupied molecular orbital (LUMO) levels undergo preferential reduction at the Cu surface, forming a mechanically protective layer. This pre-SEI effectively guides uniform Li nucleation, suppresses dendritic growth, and dramatically improves cycling stability. In contrast, when no pre-SEI is present, dendritic Li and dead Li accumulation occur even with LiF- and Li_3N-rich SEI layers, resulting in rapid capacity fading. Our findings uncover the hidden interfacial chemistry that governs the reversibility of AFLMBs and establish pre-SEI formation as a critical design principle for achieving durable, high-performance anode-free batteries.

1 | Introduction

As the demand for higher energy density in lithium-ion batteries increases, various next-generation systems have been proposed, including lithium–sulfur batteries, lithium metal batteries (LMBs), and all-solid-state batteries (ASSBs) [1–3]. Most of these systems replace the conventional graphite anode with lithium metal to enhance energy density. Anode-free lithium metal batteries (AFLMBs) represent a further advanced battery system. In AFLMBs, there is no active anode material at the beginning of cycling. Instead, lithium ions from the cathode are plated onto a current collector (typically copper) during charging, and stripped during discharge [4]. Lithium metal acts as the anode during cycling. AFLMBs offer not only the inherent advantages of lithium metal batteries such as high energy density, but also reduce manufacturing costs and complexity by eliminating the need for anode materials [5]. Furthermore, by removing the space typically occupied by the anode, more room becomes available for cathode materials, allowing AFLMBs to theoretically achieve even higher energy densities than conventional LMBs.

Despite these advantages, AFLMBs inherit critical challenges associated with lithium metal, such as dendritic growth and dead lithium formation [4, 6, 7]. Dead lithium forms when dendritic

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lithium becomes electrically isolated from the current collector and floats in the electrolyte, no longer contributing to capacity [8]. Because AFLMBs lack excess lithium, even small losses of active lithium rapidly degrade reversibility and lead to severe capacity fading. Thus, achieving uniform lithium deposition is more critical than LMBs for stable cycle performance.

Several strategies have been proposed to improve lithium deposition behavior. (1) Electrolyte engineering aims to form an inorganic-rich SEI that suppresses dendritic growth [9, 10]. Methods such as high-concentration electrolytes, dual-salt formulations, and anion-adsorbing interfacial layers have been explored to promote the formation of robust, anion-derived SEI layers [11–13]. These SEI layers enable more uniform lithium plating and enhance cycling stability. (2) Current collector modification is another approach. For example, lithophilic coatings (e.g., silver) can promote uniform Li nucleation [14, 15], and 3D structured current collectors are designed to accommodate lithium growth internally, thereby reducing dendrite formation [16, 17]. Although current collector modifications successfully regulate lithium deposition morphology, the formation of SEI layer on Li metal is inevitable and significantly influences subsequent Li deposition. Therefore, understanding and controlling SEI formation is fundamentally important for stable cycling of AFLMBs.

The SEI layer typically covers the lithium surface; therefore, previous studies have mainly analyzed the Li-metal surface after cycling to investigate the relationship between SEI composition and lithium growth, and to guide electrolyte design [11, 18, 19]. However, little attention has been paid to the interfacial reactions between electrolytes and the Cu current collector. This is because the Fermi level of Cu generally lies well below the lowest unoccupied molecular orbital (LUMO) level of electrolytes, making interfacial reduction reactions thermodynamically unfavorable and thus preventing the formation of pre-solid electrolyte interphase (pre-SEI) on the Cu surface, as shown in Figure 1a.

Nevertheless, the Cu surface plays a critical role as the initial site for lithium nucleation and deposition in AFLMBs. In this study, we investigated the possibility of pre-SEI formation on Cu current collector and its effect on subsequent Li plating. Although direct electrolyte reduction on Cu is thermodynamically unfavorable under typical conditions, the LUMO level of an electrolyte can be modulated by salt composition. By adjusting the electrolyte formulation (e.g., changing the salt chemistry), the LUMO can be sufficiently lowered, enabling interfacial reduction reactions on the Cu surface and leading to the formation of a pre-SEI (Figure 1a). In effect, this approach allows us to engineer a favorable pre-SEI on Cu that can influence subsequent Li deposition. Our findings firstly reveal that electrolyte salt chemistry dictates the formation and characteristics of the pre-SEI layer, which in turn governs the initial lithium deposition behavior on Cu. Since the initial Li plating morphology strongly impacts the final deposition structure, it ultimately governs the reversibility and performance of AFLMBs. This study highlights the previously overlooked importance of the pre-SEI formed on Cu surface and provides new design principles for tailoring electrolyte formulations to enhance performance of AFLMBs.

2 | Results

2.1 | Pre-SEI Layer Formation on Cu via Physical Contact

As localized high concentration electrolytes (LHCE), 1.125 M salt was dissolved in tetraethylene glycol dimethyl ether (TEGDME):1,1,2,2-tetrafluoroethyl-2,2,3,3-tetrafluoropropyl ether (TTE) (3:5 vol%). To regulate electrolyte composition, four different salt combinations were used with lithium bis(fluorosulfonyl)imide (LiFSI), $LiNO_3$, and lithium difluoro(oxalate)borate (LiDFOB): 1) 3 M LiFSI, 2) 2 M LiFSI and 1 M $LiNO_3$, 3) 1 M LiFSI and 2 M $LiNO_3$, and 4) 1 M LiFSI and 2 M LiDFOB. The molar concentration represents in TEGDME only except TTE. The salts have different LUMO levels respectively, known for low LUMO level of LiDFOB and $LiNO_3$ (Figure 1a) [20, 21]. These electrolytes are hereafter referred to as 3F, 2F1N, 1F2N, and 1F2D, respectively.

Anode-free LFP|Cu cells were assembled using these electrolytes, and rested for 4 h. After resting, the cells were disassembled, and then Cu surfaces were analyzed by X-ray photoelectron spectroscopy (XPS), as shown in Figure 1b. Although the electrolytes were only in physical contact with the Cu current collector (without applied current), noticeable changes were observed on the Cu surface (Figure 1c–e). These interfacial changes varied significantly depending on electrolyte composition. Notably, the 3F electrolyte exhibited no detectable surface modification and was nearly identical to pristine Cu foil. In contrast, the other electrolytes clearly induced the formation of an interfacial layer referred to as the pre-SEI layer.

Comparison of the pre-SEI layer compositions revealed that LiF is present in all three samples: 2F1N, 1F2N, and 1F2D [22, 23]. According to the F 1s spectra (Figure 1c), the 1F2N electrolyte produces a pre-SEI layer with the highest LiF content, while the 1F2D electrolyte exhibits a markedly lower LiF signal compared to the high intensity of S–F (or B–F) related species. The N 1s spectra (Figure 1d) further show that the pre-SEI layers in 2F1N and 1F2N contain various nitrogen-related species, such as Li_3N and Li_xNO_y, along with a prominent N–S bonding peak [24, 25]. In contrast, the pre-SEI formed in 1F2D exhibits negligible Li_3N content and is dominated by N–S bonding species. This difference is attributed to the nitrogen source. In 1F2D, nitrogen originates solely from the decomposition of the FSI^-, while in 1F2N and 2F1N, $LiNO_3$ provides an additional nitrogen source. Boron-containing species were identified in the B 1s spectra of the 1F2D sample, confirming the decomposition of LiDFOB as a contributor to pre-SEI formation (Figure S1). Moreover, the C 1s spectra (Figure 1e) reveal a relatively higher amount of organic compound, as well as additional Li_2CO_3 formation in 1F2D compared to the others [26]. The presence or absence of the interfacial pre-SEI layer was schematically illustrated as a function of the electrolyte composition in Figure 1f. Also, the chemical composition of the pre-SEI layer was found to depend on the electrolyte composition.

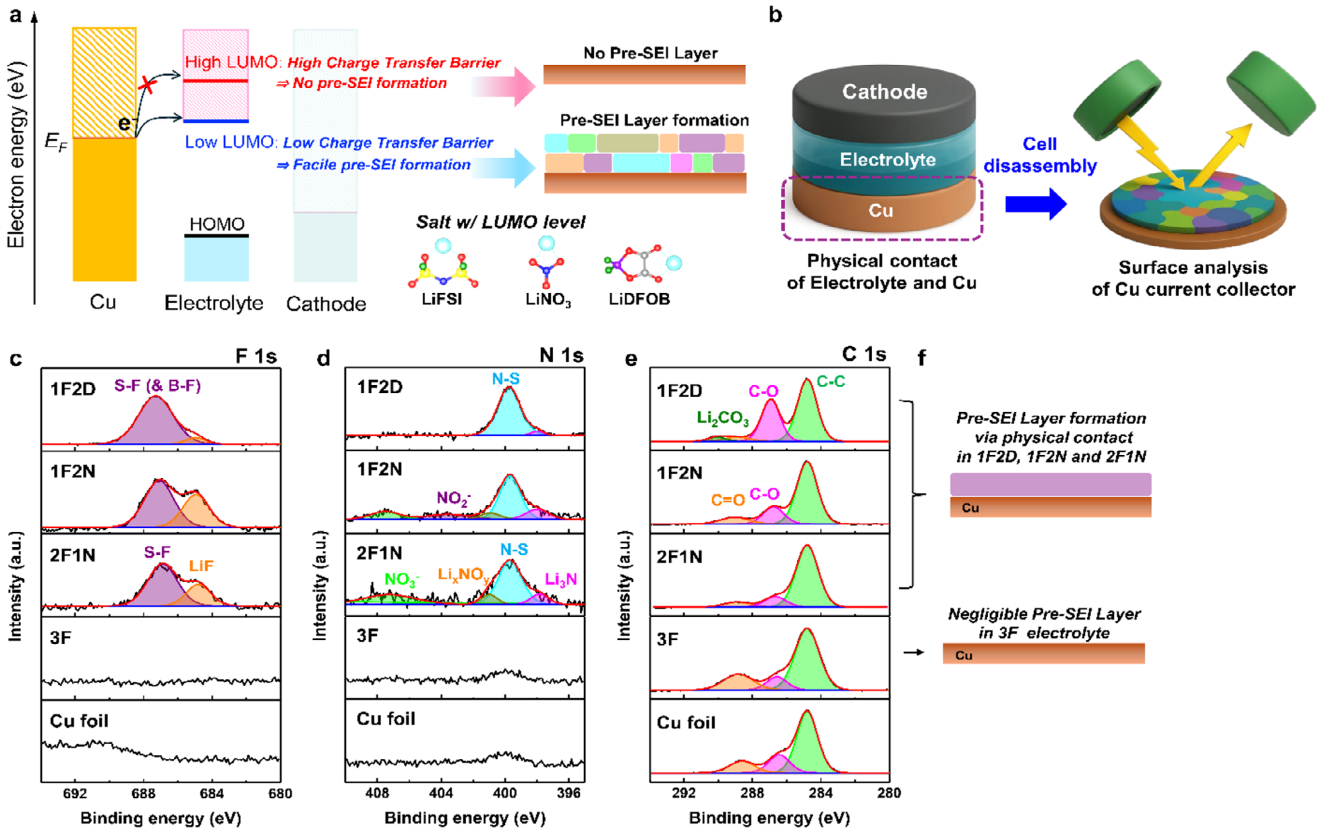

FIGURE 1 | (a) Schematic illustration of the energy-level alignment between Cu and the LUMO levels of the electrolytes, determining the formation of the pre-SEI layer. (b) Schematic illustration of the XPS analysis procedure for the Cu surface after cell rest. XPS spectra of the Cu foil after 4 h of cell rest: (c) F 1s, (d) N 1s, and (e) C 1s, (f) Schematic illustration of the formation of pre-SEI layers depending on electrolytes.

2.2 | Pre-SEI Formation Derived by Low LUMO Levels of Molecules

As all electrolytes used in this study are LHCEs, various molecule structures such as contact-ion pairs (CIPs), aggregates (AGGs), and dual-ion pair AGGs (DIP-AGGs) are expected to coexist, as illustrated in Figure 2a. AGGs are generally known to possess lower LUMO levels than free anions or CIPs [27, 28], which increases the likelihood of electrolyte reduction via electron transfer from the Cu surface. To evaluate this effect, density functional theory (DFT) calculations were performed to determine the LUMO/highest occupied molecular orbital (HOMO) levels of representative molecular structures, including CIPs, AGGs and DIP AGGs (Figure 2b) Among the three CIPs, Li^+-$DFOB^-$ exhibited the lowest LUMO level, consistent with previous reports [20, 21, 29]. Notably, the formation of AGGs leads to a decrease in the LUMO levels for all three cases ($2Li^+$-$2FSI^-$, $2Li^+$-$2NO_3^-$, and $2Li^+$-$2DFOB^-$) compared to their corresponding CIPs. Furthermore, the DIP-AGG structures, coordinated by FSI^- and NO_3^- or $DFOB^-$, exhibit an even more pronounced reduction in LUMO level. This trend implies that DIP-AGG formation markedly enhances the electrolyte's reducibility, thereby facilitating pre-SEI formation on the Cu surface. The experimentally measured Fermi level of Cu, obtained via ultraviolet photoelectron spectroscopy (UPS) [30], was approximately -4.26 eV versus vacuum (Figure S2), which is similar as the value (-4.70 eV) calculated by DFT. These values provide a reliable reference for comparing the LUMO levels of electrolyte species. Although the calculated LUMO levels remain above the Cu Fermi level, the presence of more complex or dynamic AGG species can lead to spontaneous electrolyte reduction.

Molecular structures in each electrolyte were further investigated using Raman spectroscopy (Figure 2c–f). In 3F electrolyte (Figure 2c), the S–N–S stretching vibration mode exhibits the coexistence of free FSI anions, CIPs, and AGGs. There are two distinct peaks that can be deconvoluted as AGG-I and AGG-II. Higher Raman shift peak (designated AGG-II) is frequently observed in high concentration electrolytes due to more stabilized complex AGGs, which are possibly anions coordinated by more Li cations or larger cluster with lowering energy than AGG-I [31]. In 2F1N and 1F2N of Figure 2d,e, a new peak at 710 cm^{-1} appears, which corresponds to the in-plane deformation mode of NO_3^- [32, 33]. As the $LiNO_3$ content in electrolyte increases, the intensity ratio of AGG (I and II) to CIP increases. A similar trend was observed in the Raman spectra corresponding to the symmetric stretching mode of NO_3^- (Figure S3) Here, AGG-II also possibly exists in another configuration such as DIP-AGGs, in which both FSI^- and NO_3^- are simultaneously coordinated with Li^+, as proposed in Figure 2a. Because $LiNO_3$ exhibits strong binding energy [34], dual-ion coordinated species containing FSI^- are expected to be more stabilized, thereby showing Raman shifts at higher wavenumber. Indeed, the peak of AGG-II is broader and extends toward higher wavenumbers compared to that in the 3F electrolyte.

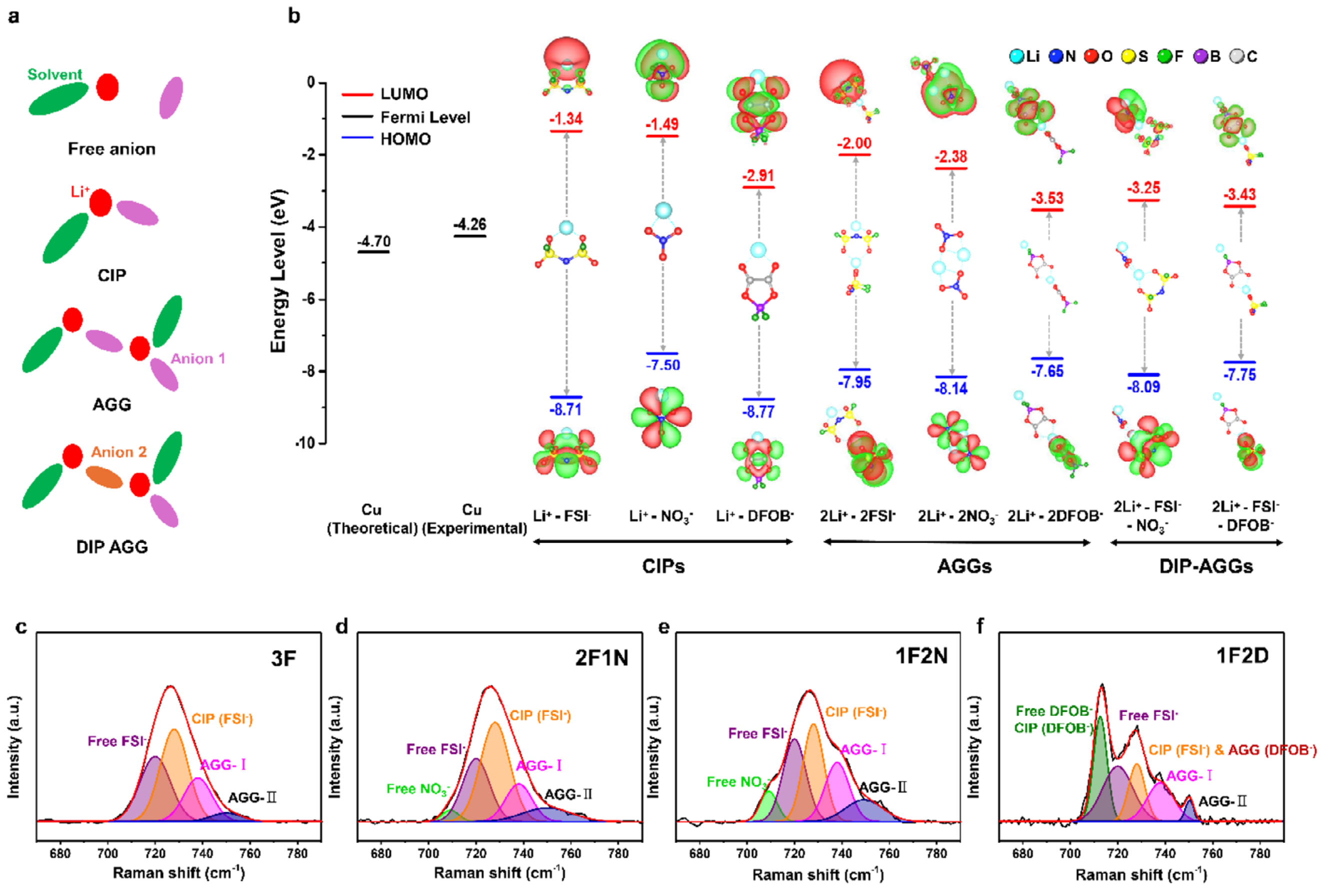

FIGURE 2 | (a) Schematic illustrations of solvation structures. (b) Energy level comparison between the Cu Fermi level and the DFT-calculated LUMO and HOMO levels of possible molecule structures in the electrolytes. Raman spectra in the range of 670–790 cm^{-1} of (c) 3F, (d) 2F1N, (e) 2F1D, and (f) 1F2D electrolytes.

In 1F2D electrolyte (Figure 2f), the O–B–O deformation mode peak originating from $DFOB^-$ indicates coexistence of both free anions and CIPs [31]. Also, $DFOB^-$-based AGG exists. However, the exact contributions could not be resolved due to spectral overlap between $DFOB^-$-based AGG and FSI^--based CIP. Nevertheless, the higher concentration of LiDFOB than LiFSI in the electrolyte implies a higher proportion of $DFOB^-$-based AGGs, which was further supported by the decreased intensity observed upon reducing the LiDFOB salt content (Figure S4). Additionally, AGG-I and II peaks were also identified. AGG-I is likely to be associated with DIP-AGGs containing $DFOB^-$. Unlike NO_3^-, which has strong binding energy, $DFOB^-$ exhibits weaker coordination strength [35]; thus, its DIP-AGGs may display vibration modes with lower energy. Furthermore, AGG-II peak shows relatively low intensity and a narrow bandwidth.

In order to examine the aggregation behavior of Li^+ ions and various anions, molecular dynamics (MD) simulations were performed (Figure S5). The MD results qualitatively support the Raman-spectroscopic identification of CIPs and AGGs. Analysis of the MD trajectories shows that Li^+ ions frequently coordinate directly with a single anion (FSI^- or NO_3^-/$DFOB^-$), forming CIPs, and also participate in localized multi-ion clusters characteristic of AGGs, consistent with the speciation inferred from Raman spectra. These coordination motifs are further supported by the radial distribution functions averaged over the simulation time. Together, these results indicate that the dual-salt electrolyte promotes pronounced ion pairing and aggregation, and the MD simulations provide atomistic insight into the local ionic environments that complement the experimental observations.

However, caution is warranted in interpreting these results quantitatively, as the finite simulation cell size and limited sampling time inherently constrain the spatial and temporal resolution of ionic configurations. As a result, the relative populations and detailed structures of CIPs and AGGs captured in the simulations may not fully reflect those in the bulk electrolyte inferred from Raman measurements. Nevertheless, the MD results robustly corroborate the presence of CIPs and AGGs and elucidate their local structural characteristics, thereby supporting the experimental findings at the molecular level.

These results confirm the coexistence of various AGG structures across all electrolytes. Importantly, the AGGs content increases significantly in dual-salt systems compared to the single-salt 3F electrolyte. When combined with the DFT results, the formation of DIP-AGGs with lower LUMO levels in dual-salt electrolytes (2F1N, 1F2N, and 1F2D) facilitates electrolyte reduction at the Cu surface during the resting period, promoting the formation of a pre-SEI layer. This finding highlights the critical influence of salt composition and molecular speciation on interfacial stability of Cu in AFLMBs.

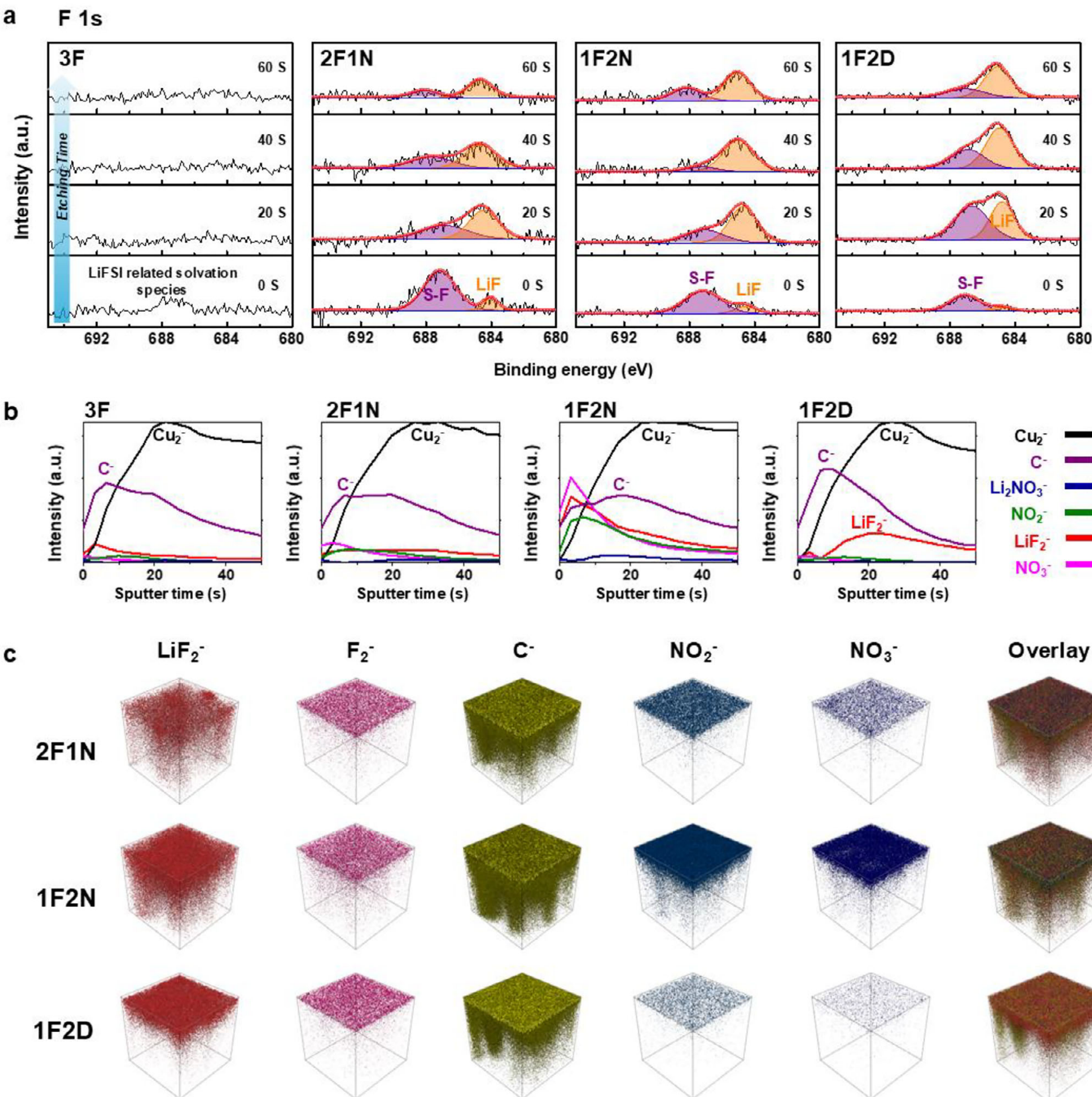

FIGURE 3 | (a) XPS depth profiles of F 1S spectra of the pre-SEI layer formed in different electrolytes (b) ToF-SIMS depth profiles of various chemical species as function of sputter time. The intensities are normalized to their respective maximum value. (c) 3D rendering for chemical species and their overlay, illustrating the spatial distribution of the pre-SEI layer.

2.3 | Electrolyte-Dependent Composition of the Pre-SEI Layer

Since AGGs with low LUMO levels can induce pre-SEI formation on the Cu surface through physical contact, the composition of the pre-SEI layer is highly dependent on the salt composition, shown in Figure 1c–e. To gain deeper insight into the structure and vertical composition of the pre-SEI layer, depth profile XPS and time-of-flight secondary ion mass spectrometry (ToF-SIMS) analysis were performed.

For the 3F electrolyte, only weak signals were observed in both XPS and ToF-SIMS analyses (Figure 3 and Figure S6), indicating a negligible pre-SEI layer. In contrast, a distinct pre-SEI layer was formed in the other electrolytes, 2F1N, 1F2N, and 1F2D. Upon increasing the $LiNO_3$ salt content while decreasing LiFSI (i.e., comparing 2F1N and 1F2N), both S–F and Li–F peaks were observed in the F 1s XPS spectra (Figure 3a). As the etching time increased, the intensity of the S–F peak decreased, while that of the Li–F peak increased in both electrolytes, indicating progressive enrichment of LiF toward the inner region of the pre-SEI layer.

For the 1F2D electrolyte, this compositional evolution was more pronounced. The LiF component was negligible at the outer surface, while both S–F and Li–F signals increased with etching time (Figure 3a), corresponding to the inner pre-SEI region. Notably, the LiF content became dominant over S–F species in the inner layer. These trends are consistent with the ToF-SIMS results shown in Figure 3b, where LiF_2^- signals were weak at the outermost surface but became prominent after 20 s of sputtering, indicating that LiF-based inorganic species are primarily concentrated in the inner pre-SEI region. In contrast, relatively strong carbon-related signals were detected near the surface, suggesting that the outer pre-SEI layer is dominated by organic components.

Compared to 1F2D, the 1F2N electrolyte exhibited a lower carbon-related species even at the outer surface (0 s sputtering in Figure 3b), accompanied by higher intensities of NO_3^- and LiF_2^- signals up to 10 s of sputtering. The simultaneous detection of NO_3^-, NO_2^-, and $Li_2NO_3^-$ further indicates that $LiNO_3$-derived species actively contribute to pre-SEI formation. When the $LiNO_3$ content was reduced (2F1N electrolyte), the outer pre-SEI region showed a higher carbon fraction and reduced nitrogen-related and LiF components.

Overall, these XPS and ToF-SIMS results demonstrate that not only the presence but also the vertical distribution and chemical composition of the pre-SEI layer are strongly dependent on the electrolyte formulation. The ToF-SIMS 3D distribution images in Figure 3c further support this conclusion, revealing differences in pre-SEI thickness among the electrolytes. Among the samples, 2F1N exhibited the lowest overall signal intensity, consistent with the thinnest pre-SEI layer, as also confirmed by the XPS depth profile results in Figure S6. At an etching time of 60 s, the Cu–O peak intensity was highest for 2F1N among the pre-SEI-forming electrolytes, whereas the lowest Cu–O signal was observed for 1F2D, indicating the formation of the thickest pre-SEI layer. As a result, the electrolyte-dependent differences in pre-SEI composition and thickness are expected to play a critical role in regulating the initial lithium deposition and growth behavior.

2.4 | Pre-SEI Layer Impacts on Li Deposition Morphology

The impacts of pre-SEI layer on Li deposition were investigated in anode-free LFP|Cu cells with each electrolyte by scanning electron microscope (SEM). Li deposition morphology on Cu was observed at state of charge (SOC) 25% and 100% in Figure 4a. At SOC 25%, that is early stage of Li deposition, distinct differences of morphologies were observed. In the 3F electrolyte, where no pre-SEI is formed, dendritic Li morphology observed, growing predominantly toward the separator rather than spreading laterally along the surface Such vertical deposition is likely to transform into dead lithium during subsequent cycles, leading to capacity loss and reduced coulombic efficiency. In contrast, in the electrolytes that induce pre-SEI formation (1F2D, 2F1N, and 1F2N), Li deposition exhibited comparatively lateral growth morphology, spreading along the surface rather than vertically toward the separator. Although dendritic features are still observed in the 1F2D electrolyte, the deposited lithium appears noticeably flatter than that formed in the 3F electrolyte. This behavior suggests that the presence of the pre-formed SEI guides planar lithium deposition, leading to more uniform and stable Li plating.

As lithium deposition proceeds to SOC 100%, these differences become more pronounced. The 3F exhibits severely dendritic lithium deposition with strong vertical growth. In the 1F2D, laterally deposited lithium is partially observed; however, a large number of needle-like dendritic lithium structures remain, indicating non-uniform lithium deposition. In contrast, the 2F1N and 1F2N electrolytes maintain dense, and lateral lithium morphologies. Among them, the 1F2N electrolyte shows the highest uniformity with fewer voids compared to the 2F1N electrolyte. Consistent trends were also observed in lithium stripping morphology, showing similar electrolyte-dependent features to those of lithium deposition (Figure S7).

To quantitatively validate the morphological trends observed by SEM, atomic force microscopy (AFM) was employed to analyze the surface roughness of deposited lithium after the first charging step in LFP|Cu cells (Figure 4b). The surface roughness was quantified using the arithmetic average roughness (Ra) extracted from line profiles (Figure S8). The 3F electrolyte exhibited the highest surface roughness, with a Ra value of 0.469 µm, consistent with the highly dendritic and vertically grown lithium morphology observed in Figure 4a. The 1F2D electrolyte showed a moderately reduced roughness (Ra = 0.262 µm), reflecting partial lateral lithium growth but with remaining dendritic features. In contrast, significantly smoother lithium surfaces were obtained for the 2F1N and 1F2N electrolytes, with Ra values of 0.200 and 0.179 µm, respectively. These AFM results quantitatively support the SEM observations, confirming that lithium deposition becomes increasingly lateral and uniform in electrolytes forming a pre-SEI layer.

Lithium deposition morphology and surface roughness clearly varied depending on the electrolyte. To understand the origin of these differences, the lithium deposition behavior was further interpreted in terms of lithium nucleation and growth mechanisms [36]. Chronoamperometry (CA) measurements were conducted at a fixed overpotential of −250 mV (Figure S9). The obtained current–time transients were normalized into the forms of $(j/j_m)^2$ versus (t/t_m), where j_m and t_m represent the maximum current density and the corresponding time, respectively, and then compared with classical nucleation models to distinguish between instantaneous and progressive nucleation as well as 2D and 3D growth behaviors [37].

$$\text{3DI: } (J/J_{\max})^2 = 1.9542(t/t_{\max})^{-1}[1-\exp(-1.2564(t/t_{\max}))]^2$$

$$\text{3DP: } (J/J_{\max})^2 = 1.2254(t/t_{\max})^{-1}\left[1-\exp\left(-2.3367(t/t_{\max})^2\right)\right]^2$$

$$\text{2DI: } (J/J_{\max})^2 = (t/t_{\max})^2\left(\exp\left\{\frac{1}{2}\left[1-(t/t_{\max})^2\right]\right\}\right)^2$$

$$\text{2DP: } (J/J_{\max})^2 = (t/t_{\max})^4\left(\exp\left\{\frac{2}{3}\left[1-(t/t_{\max})^3\right]\right\}\right)^2$$

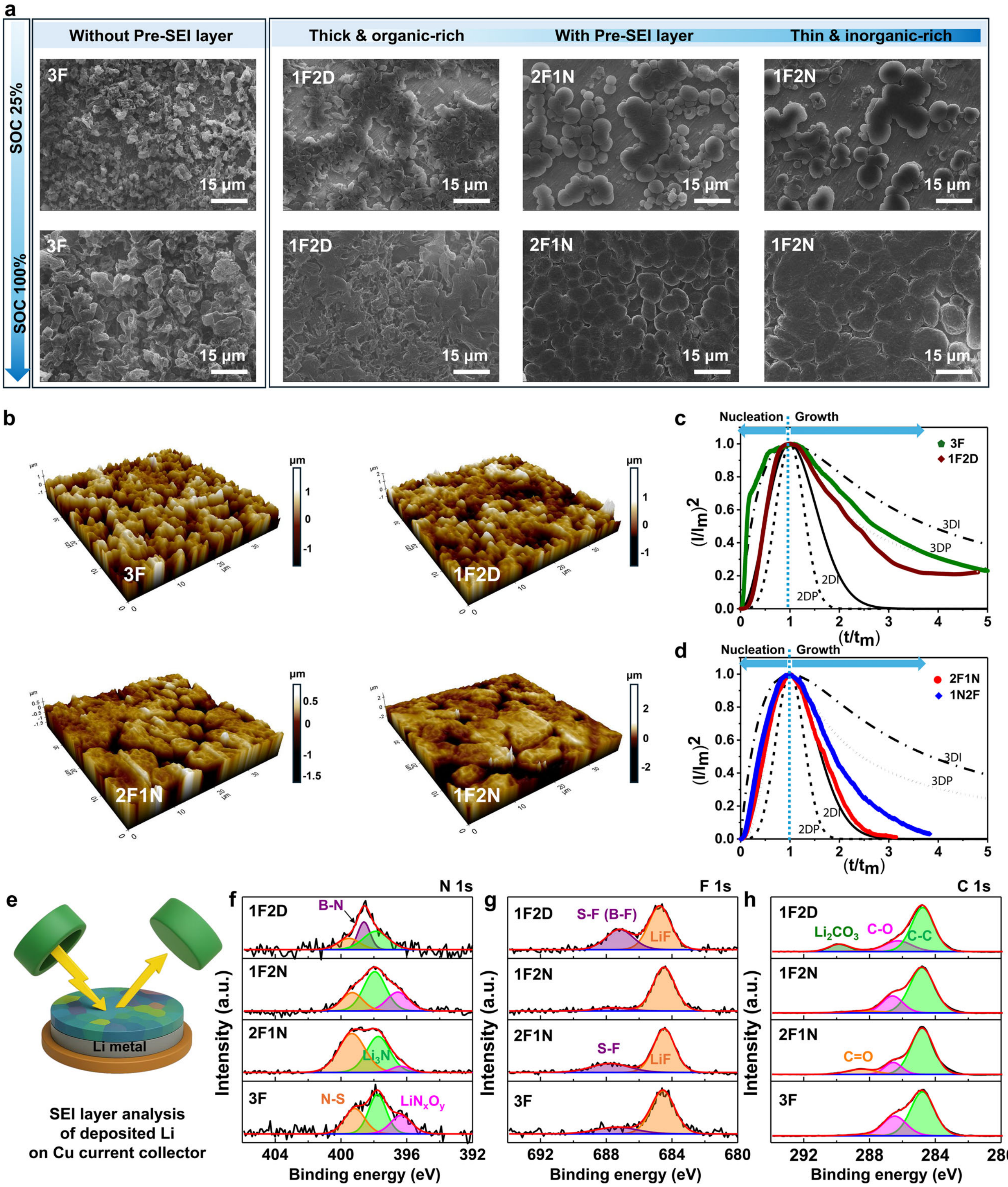

FIGURE 4 | (a) SEM images of lithium deposition morphologies on Cu at SOC 25% and SOC 100% in LFP|Cu cells with the electrolytes. (b) 3D AFM images of plated lithium after first charging in LFP|Cu cells. Dimensionless current-time transients in comparison with the classical 2D and 3D nucleation models in (c) 3F and 1F2D, (d) 2F1N and 1F2N. (e) Schematic illustration of SEI analysis on deposited Li on Cu (after charging) by XPS. XPS spectra of the SEI layer after the first charging in LFP|Cu cells with different electrolytes: (f) N 1s, (g) F 1s, and (h) C 1s.

The current–time transients for the 2F1N and 1F2N electrolytes closely overlap with the 2D instantaneous (2DI) nucleation model in Figure 4d, indicating that lithium deposition in both systems is primarily governed by rapid 2D nucleation followed by lateral growth. The negligible contribution from the 3D progressive (3DP) component, evidenced by the absence of a pronounced long-tail behavior, suggests that vertical 3D growth is effectively suppressed.

In these systems, lithium initially spreads laterally across the Cu surface to form a continuous 2D layer, and subsequent lithium nuclei preferentially form and grow on the pre-deposited lithium in a predominantly 2D manner, resulting in a dense and laterally extended lithium morphology. In contrast, the experimental current–time transients for the 3F in Figure 4c deviate markedly from the 2DI model and exhibit significant contributions from 3D growth mechanisms. The pronounced long-tail behavior following the current maximum indicates that lithium deposition in these electrolytes is dominated by vertical 3D and dendritic growth rather than uniform lateral spreading. In the case of the 1F2D electrolyte (Figure 4c), the experimental current transient closely follows the 2DI nucleation model at the initial stage, indicating that lithium nucleation initially proceeds in a predominantly 2D manner. This behavior is clearly distinct from that of the 3F electrolyte, where 3D growth dominates from the beginning. However, as deposition proceeds, the contribution from the 3DP mechanism becomes increasingly significant in 1F2D, leading to a transition toward 3D lithium growth at later stages.

Importantly, in electrolytes forming a pre-SEI layer (2F1N, 1F2N and 1F2D), the experimental current–time transients show close agreement with the 2DI nucleation model prior to the current maximum, corresponding to the lithium nucleation stage. This indicates that the pre-SEI layer provides a more homogeneous interfacial environment on Cu, thereby reducing spatial variations in nucleation energy barriers and favoring 2DI nucleation. Beyond the current maximum, where lithium growth becomes dominant, the growth behavior becomes increasingly dependent on the composition of the pre-SEI layer. When the pre-SEI layer contains highly Li-ion-conductive components such as NO_x species, lateral lithium growth is favored, as observed for the 1F2N electrolyte, whereas the 1F2D electrolyte, which contains negligible NO_x species and forms a relatively thick pre-SEI layer, follows predominantly 3DP lithium growth due to limited Li^+ transport. These results suggest that pre-SEI formation primarily governs the lithium nucleation process, whereas the subsequent lithium growth behavior is regulated by the chemical composition of the pre-SEI layer.

As lithium is deposited on the Cu current collector, the pre-SEI layer gradually evolves into a SEI layer through the participation of lithium in SEI formation. The evolution of interfacial resistance during lithium deposition was investigated by electrochemical impedance spectroscopy (EIS) measurements of Li|Cu cells (Figure S10). In the 1F2N and 2F1N electrolytes, the interfacial resistance remained nearly constant regardless of the SOC, while only the charge-transfer resistance decreased during cycling.

In contrast, for the 1F2D electrolyte, the interfacial resistance gradually decreased, likely due to lithium enrichment within the pre-SEI layer, which enhances Li^+ transport. However, it should be noted that the resistance is significantly higher than that of the other samples at SOC 25%, based on equivalent circuit analysis (Figure S11). This high resistance is attributed to the thick pre-SEI layer and the absence of NO_x-derived species. For the 3F electrolyte, which does not form a pre-SEI layer, the SEI resistance gradually decreases as lithium deposition proceeds. Interestingly, the charge-transfer resistance in the 3F system is markedly lower than that of the other electrolytes (Table S2). This can be attributed to the dendritic and vertically aligned lithium growth, which significantly increases the electrochemically active surface area and thereby reduces the charge-transfer resistance.

After full charging (i.e., complete lithium deposition), the compositions of the SEI layers were analyzed by XPS to examine their differences from the pre-SEI layers (Figure 4e). Because a larger amount of lithium participates in SEI formation after charging, the intensities of lithium-containing species such as LiF and Li_3N increased compared to those in the pre-SEI layer. In the F 1s spectra (Figure 4g), the highest LiF/S–F ratio was observed for the 1F2N electrolyte and the lowest for 1F2D. In N 1s spectra (Figure 4f), nitrogen-containing species such as N–S, N–B [38], and Li_3N were detected in the SEI formed with the 1F2D electrolyte. In both the 1F2N and 3F electrolytes, Li_3N was the predominant nitrogen species, whereas in the 2F1N sample, comparable intensities of N–S and Li_3N were observed. Although the 3F electrolyte does not form a pre-SEI layer, it should be noted that a LiF- and Li_3N-rich SEI is formed after charging.

In C 1s spectra (Figure 4h), Li_2CO_3 was detected in the SEI of the 1F2D electrolyte, showing a similar trend to that observed in the pre-SEI composition. Because Li_2CO_3 can spontaneously react with Li to form Li_2O, it is considered an undesirable component for Li protection [39]. Indeed, a larger amount of Li_2O was detected in the SEI layer formed in the 1F2D electrolyte (Figure S12). These differences in SEI composition are expected to influence the reversibility and cycling stability of AFLMBs.

2.5 | Reversibility of Anode-Free Li Metal Batteries

Pre-SEI layer strongly affects Li morphology at the initial deposition. SEI which is formed after Li deposition has different chemistry due to different salt compositions. Both factors are expected to affect the reversibility of AFLMBs. Reversibility was confirmed firstly using Li|Cu cells. The Aurbach method [40] was used to evaluate the reversibility in Figure 5a. An initial plating of 3 mAh cm^{-2} of Li was carried out at 1 mA cm^{-2}, followed by 50 cycles of plating/stripping with a capacity of 1 mAh cm^{-2}. After cycles, all residual lithium was stripped to quantify irreversible lithium loss (Reversible capacity in Figure 5b). Since 3 mAh cm^{-2} of Li was deposited at the beginning, 3 mAh cm^{-2} of capacity should be obtained theoretically. Ratios of the stripped capacity at 50^{th} to first plating capacity (Q_{S50th}/Q_{P1st}) were calculated; 35.4, 62.5, and 76.6% were obtained for 3F, 2F1N, and 1F2N respectively. 1F2N exhibits the best reversibility, whereas 1F2D exhibited a rapid increase in overpotential and early cell failure.

Because Li plating behavior strongly influences reversibility, Li nucleation overpotential was also examined (Figure 5c). The electrolyte without a pre-SEI (3F) showed the lowest nucleation overpotential (42 mV), while electrolytes with a pre-SEI showed higher values (2F1N: 138.5 mV, 1F2N: 113.4 mV, 1F2D: 200.7 mV). This is attributed to the additional energy barrier encountered when Li^+ passes through pre-SEI to nucleate. Notably, ionically conductive pre-SEI such as those in1F2N and 2F1N tended to show lower nucleation overpotentials. The voltage profiles during

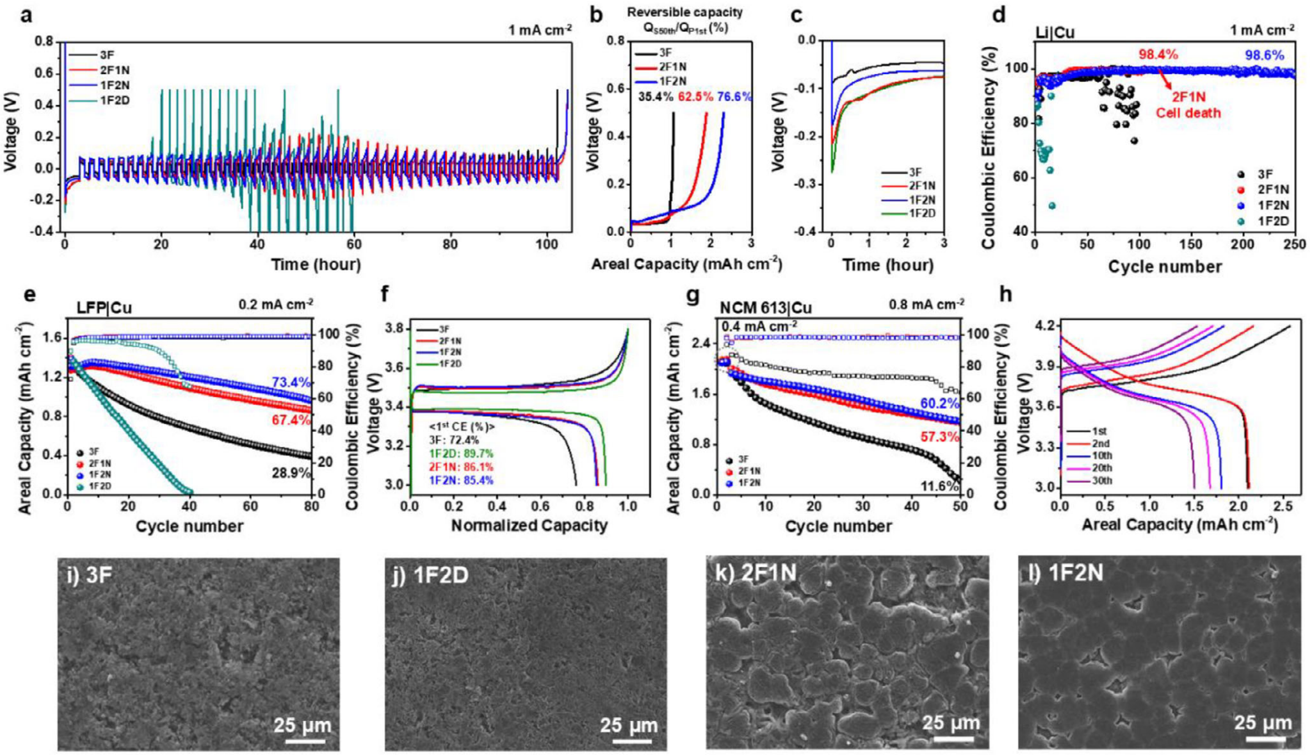

FIGURE 5 | (a) Voltage–time profiles of Li|Cu cells by the Aurbach method using different electrolytes. (b)Reversible capacities at the 50th cycle from (a). (c) Li nucleation overpotentials during the first Li deposition in (a). (d) Coulombic efficiencies of Li|Cu cells during long-term cycling (e) Capacity retention of LFP|Cu cells and their (f) Voltage profiles at the first cycle. (g) Cycling performance of NCM613|Cu cells and (h) its voltage profiles for the 1F2N electrolyte. SEM images of deposited Li morphologies in LFP|Cu cells with (i) 3F, (j) 1F2D, (k) 2F1N and (l) 1F2N electrolyte after 20 cycles.

repeated Li plating/stripping further highlight reversibility. As shown above, 3F showed a low cell overpotential due to its very low charge-transfer resistance. This, however, corresponds to high surface area due to the vertically dendritic Li growth rather than laterally uniform plating, which resulting in the poor reversibility (Figure 5b). In both 2F1N and 1F2N, the overpotential increased but subsequently stabilized upon cycling, with more pronounced variation in 2F1N. Although the origin of this difference remains unclear, the smaller overpotential fluctuation in 1F2N is likely associated with the higher contents of LiF and Li_3N in its SEI layer (Figure 4f,g).

Long-term cycling stability was further assessed in Li|Cu cells with repeated plating/stripping of 1 mAh cm^{-2} (Figure 5d). The 1F2D showed a rapid decline in coulombic efficiency before reaching 20 cycles while 3F exhibited longer cycle stability up to ≈100 cycles. It is likely due to the high resistance of SEI layer in 1F2D electrolyte (Figure S10), which possibly accelerates the irreversibility. The 2F1N electrolyte delivered improved stability compared to 3F but eventually experienced cell short-circuiting beyond 100 cycles. By contrast, the 1F2N electrolyte maintained an average coulombic efficiency of 98.6% over 250 cycles.

To evaluate practical applicability, anode-free full cells were assembled with high-mass-loading (10–12 mg cm^{-2}) $LiFePO_4$ (LFP) as a cathode and Cu current collectors, and cycled at 0.2 mA cm^{-2} (Figure 5e). The 1F2D cell exhibited the most severe degradation among all electrolytes. Its capacity rapidly decreased and was completely depleted within only 40 cycles, which is consistent with the Li|Cu half-cell results. Also, the coulombic efficiency drastically reduced in the 1F2D electrolyte. The 3F shows better reversibility, resulting in only 28.9% capacity retention after 80 cycles. 1F2N and 2F1N exhibited markedly improved electrochemical performance. The 1F2N and 2F1N cells maintained much higher stability, with 1F2N achieving the best performance, retaining 73.4% of its capacity, compared to 67.4% for 2F1N.

The first cycle voltage profiles of the LFP|Cu full cells are presented in Figure 5f. The initial coulombic efficiency (ICE) was very low in 3F (72.4%), but increased in 2F1N (86.1%), 1F2N (85.4%), and 1F2D (89.7%). In the presence of pre-SEI, 2F1N, 1F2N and 1F2D electrolytes exhibited higher ICEs, can mitigate Li loss originating from dead Li. Unexpectedly, despite its poor reversibility, 1F2D showed both the smallest voltage hysteresis and the highest ICE. It might be because of electrolyte compatibility with LFP. When LFP|Li half cells with each electrolyte were cycled (Figure S13), the 1F2D electrolyte also exhibited the lowest voltage hysteresis and the highest ICE (95.1 %), indicating its superior compatibility with LFP. However, because of the poor stability of the SEI layer, the cell degraded rapidly in later cycles.

To further verify the high voltage feasibility of the electrolytes, oxidation stability was investigated using linear sweep voltammetry (LSV) (Figure S14). All electrolytes exhibited oxidative stability above ≈4.2 V, indicating sufficient resistance to oxidative decomposition under high-voltage conditions. Notably, the 1F2D electrolyte showed the highest oxidation stability, with no pronounced decomposition current observed up to ≈ 4.8 V.

Based on the LSV results, full-cell tests were conducted using AFLMBs with a commercial $LiNi_{0.6}Co_{0.1}Mn_{0.3}O_2$ (NCM613) cathode (16 mg cm^{-2}) to evaluate practical high-voltage cell performance (Figure 5g). Although the 1F2D electrolyte exhibited superior oxidation stability in the LSV measurements, it was not further evaluated in the NCM613|Cu full-cell configuration due to its poor lithium plating/stripping behavior and low reversibility, as demonstrated in the LFP|Cu and Li|Cu cell tests. Among the evaluated electrolytes, the 3F electrolyte exhibited poor reversibility, retaining only 11% of its capacity after 50 cycles. In addition, the Coulombic efficiency continuously decreased during cycling, indicating severe irreversible lithium loss and unstable lithium plating/stripping behavior. In contrast cells with the 1F2N and 2F1N achieved significantly higher reversibility, with 60% and 57% capacity retention respectively and maintained relatively stable Coulombic efficiency throughout cycling. The representative voltage profile of 1F2N is presented in Figure 5h, while those of the other electrolytes are provided in Figure S15.

Deposited Li morphologies after 20 cycles were observed (Figure 5i–l). In 3F, crushed, porous and small particles-like Li were deposited, which is likely due to dead Li for cycling. In 1F2D, threadlike morphology appeared, however, it looks less dead Li than 3F. It might be because high resistance interface property can cause severe degradation rather than Li loss from the dead Li. The 2F1N cell exhibited comparatively smoother deposition, but local irregularities emerged with cycling due to insufficient interfacial durability. By contrast, Li in the 1F2N showed dense and uniformly deposited for cycling.

3 | Discussion

This study is the first to demonstrate that a pre-SEI can be generated on the Cu surface of AFLMBs solely through physical contact with the electrolyte, and that this pre-SEI significantly influences the subsequent lithium deposition morphology. However, the mere presence of pre-SEI does not guarantee high reversibility. Both properties of the pre-SEI and the SEI play critical roles in determining the reversibility of AFLMBs, as schematically illustrated in Figure 6.

When the molecules in the electrolyte do not possess sufficiently low LUMO levels relative to the Cu, as shown in Figure 6a, a pre-SEI layer is not formed, leading to vertically dendritic Li growth during the initial Li deposition. In contrast, when molecular structures such as dual-ion pair aggregates (DIP-AGGs) with low LUMO levels are present in Figure 6b, electrolyte reduction occurs at the Cu surface, resulting in pre-SEI formation. The composition, thickness, and vertical distribution of the pre-SEI varies depending on the salt chemistry.

When a high concentration of $LiNO_3$ is included among the salts such as 1F2N, thin and ionically conductive pre-SEI layers with inorganic species distributed across the interface. Such pre-SEIs enable more homogeneous Li^+ transport, thereby facilitating lateral both lithium nucleation and growth on the Cu surface. In contrast, although a pre-SEI can also form in LiDFOB-based electrolytes, this layer is substantially thicker and exhibits a pronounced vertical compositional gradient, with organic-rich species dominating the outer region and inorganic components concentrated primarily in the inner region. This is likely because LiDFOB exhibits relatively weak ionic association, causing Li^+ to remain more solvated by solvent molecules compared with $LiNO_3$-containing systems [34]. Although the high-resistance pre-SEI layer can initially promote lateral lithium nucleation, it cannot sustain 2D lithium growth, ultimately leading to dendritic growth. However, the resulting dendrites appear more flattened and compressed compared to those formed on Cu without a pre-SEI layer, suggesting that the pre-SEI may serve as a mechanically protective layer.

Superior reversibility and cycling stability were primarily attributed to the formation of a LiF- and Li_3N-rich SEI layer, together with a protective and ionically conductive pre-SEI. The 3F electrolyte, which lacks a pre-SEI but forms an LiF- and Li_3N-rich SEI layer, exhibited better reversibility than the LiDFOB-based electrolyte that possesses a pre-SEI yet forms a more organic-rich SEI. This finding implies that not only the formation of a pre-SEI but also the formation of a robust SEI is critically important for achieving stable cycling in AFLMBs. Nevertheless, even under conditions where both LiF- and Li_3N-rich SEI layers were present, only the cell with a pre-SEI maintained superior cycling stability, underscoring its crucial role in initial interfacial stabilization.

To further evaluate the cycling stability of the best-performing 1F2N electrolyte, we employed a small amount of Li_3N (7.5 wt%) as a sacrificial cathode [41] together with LFP (77.5 wt%). The sacrificial Li_3N was expected to compensate for lithium losses during SEI formation and thereby enhance the cycle stability. At first charge, cell delivered a capacity of ≈240 mAh g^{-1}, exceeding the theoretical capacity of LFP, which indicates successful lithium supply from Li_3N (Figure S16). Upon discharge, the capacity decreased to ≈140 mAh g^{-1} as Li was intercalated into LFP. Initial Li deposition morphology was examined to verify whether the pre-SEI strategy remains effective under this modified cathode configuration. Uniform and flattened Li deposition on Cu was observed after full charge (Figure S17), confirming that pre-SEI approach can be extended to other anode-free cells. Furthermore, remarkable high stability was achieved, maintaining 47.6% of its capacity over 200 cycles despite the absence of external pressure (Figure 6c).

By comparison of cycle performance of AFLMBs, our anode-free cell (red star in Figure 6d) exhibits cycling stability that is comparable to or better than most previously reported liquid electrolyte-based anode-free systems, despite using only a conventional Cu foil and our electrolyte design under zero applied stack pressure. Although a few studies report longer cycle life, those results rely on high external pressure and/or specially engineered current collectors, which can increase complexity and cost. Taken together, these results indicate that our approach can achieve competitive durability while maintaining a simple and scalable cell configuration. These findings demonstrate that the combination of a sacrificial lithium source and an electrolyte capable of forming a low-resistance pre-SEI is a promising strategy to improve reversibility in AFLMBs.

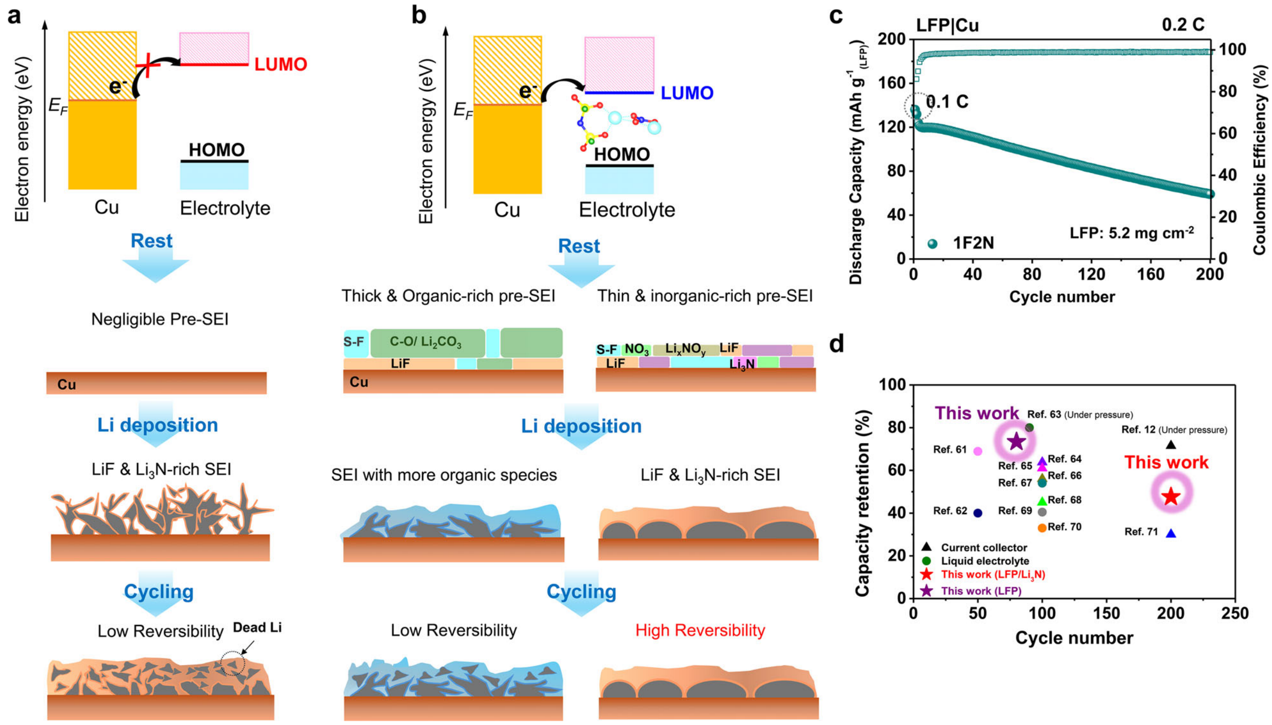

FIGURE 6 | Schematic images about this study (a) non pre-SEI formation (b) with pre-SEI formation. (c) Discharge capacity per cycle of LFP/Li_3N|Cu cell with 1F2N. (d) Cycling performance of AFBs in this study and previous other studies [12, 61–71].

4 | Conclusion

This study reveals that the pre-SEI layer, spontaneously formed on Cu through molecular aggregation in LHCEs, plays a decisive role in governing the initial Li deposition behavior and reversibility of AFLMBs. The formation of aggregates with lower LUMO levels than individual salts drives electrolyte reduction and pre-SEI formation even before electrochemical cycling. Although the pre-SEI effectively suppresses dendritic Li growth, its impact is highly dependent on its chemical composition. Ionically conductive pre-SEIs promote uniform and planar Li deposition, whereas thick and organic-dominated pre-SEIs exhibit higher interfacial resistance and facilitate dendritic growth. The salt chemistry thus dictates both pre-SEI and subsequent SEI compositions, ultimately determining interfacial stability and cycling reversibility. In particular, Li_2CO_3- and Li_2O-containing SEIs derived from organic-rich layers exhibit poor reversibility, while LiF/Li_3N-rich interfaces enable highly stable cycling. Moreover, despite the formation of LiF- and Li_3N-rich SEIs in the 3F electrolyte, the absence of a pre-SEI layer results in limited cycling stability, underscoring the critical role of the pre-SEI in achieving high cycle stable AFLMBs. Overall, this work establishes the mechanistic foundation of spontaneous pre-SEI formation and highlights interfacial chemistry design as a key strategy toward high-performance AFLMBs.

5 | Experimental Method

5.1 | Electrolyte Preparation

LiFSI (>98%) and TTE (>95%) were purchased from TGI•SEJI CI. TEGDME (>99%) and LiDFOB were obtained from Sigma-Aldrich. $LiNO_3$ (>99%, anhydrous) was purchased from Thermo scientific. For the preparation of electrolytes, 3 M of two lithium salts (LiFSI + $LiNO_3$ or LiDFOB) were dissolved in TEGDME. The solution was left to stand until the lithium salts were completely dissolved. Subsequently, TTE was added to the 3 M solution at a 3:5 v% ratio and the mixture was stirred to obtain a 1.125 M dual salt electrolyte. All electrolytes were prepared in an Ar-filled glovebox with O_2 and H_2O concentration below 0.1 ppm.

5.2 | Electrode Fabrication

The LFP cathode was prepared by slurry-casting method. LFP, acetylene black, and polyvinylidene fluoride (PVDF) were mixed and ground in a mortar for 20 min. N-methyl-2-pyrrolidone(NMP) was then added to obtain a uniform slurry. The resulting slurry was cast onto a carbon-coated aluminum current collector. The LFP/Li_3N composite cathode was fabricated by a dry-electrode process due to high reactivity of Li_3N and its poor compatibility with NMP. The Li_3N powder was ground in a mortar for 30 min before electrode fabrication. LFP, Li_3N, acetylene black, multi-walled carbon nanotubes (MWCNTs), and polytetrafluoroethylene (PTFE) were mixed at a weight ratio of 77.5:7.5:5:5:5. The conventional LFP cathode had 10–12 mg cm^{-2} of mass loading and the LFP mass loading in the LFP/Li_3N composite cathodes was 5.2 mg cm^{-2}. The NCM613 cathode with mass loading of 16 mg cm^{-2} were obtained from POSCO holdings.

5.3 | Electrochemical Characterization

The anode free coin cell (CR2032) was assembled in Ar-filled glove box with 13 mm diameter disc of cathode, a monolayer polypropylene separator (Celgard 2400), 15 mm diameter Cu foil (MTI Korea) disc and 120 μL of electrolyte. The Cu foil was treated with 1 M hydrochloric acid (HCl) in deionized water to remove native oxide. A 13 mm diameter lithium metal disc with a thickness of 20 μm was used in Li|Cu cells, and lithium metal anode with a thickness of 200 μm was used in LFP|Li half cells. All galvanostatic tests were conducted using NEWARE BTS at 30°C. The LFP|Cu, LFP/Li_3N|Cu, LFP|Li cell was cycled within 3.0–3.8 V whereas the NCM613|Cu cell test was performed within a voltage range of 3.0–4.2 V. The LFP|Cu cells were operated at a current density of 0.2 mA cm^{-2} and the NCM613|Cu cell was cycled at 0.4 mA cm^{-2} for two activation cycles and was subsequently tested at 0.8 mA cm^{-2}. The LFP/Li_3N|Cu and LFP|Li cells were cycled at 0.1 C for two activation cycles and then continued at 0.2 C (1 C = 170 mA g^{-2}). The Li|Cu cell was operated at a current density of 1 mA cm^{-2}. EIS measurements were performed over a frequency range from 1 MHz to 25 mHz. LSV tests were carried out at a scan rate of 0.1 mV s^{-1}. The EIS and LSV test were conducted using Biologic electrochemical potentiostat/galvanostat device. CA measurements were performed using Li|Cu cells at a fixed overpotential of −250 mV to analyze lithium nucleation and growth behavior.

5.4 | Characterizations

The morphological analysis of the deposited lithium was performed via SEM using an SNE-4500 M Plus. The SEM was operated at 15 kV. The XPS spectra were obtained from NEXSA instrument. All XPS spectra were calibrated using carbon-carbon peak at 284.8 eV. UPS was performed using NEXSA under ultrahigh vacuum. The spectra acquired using He I radiation ($h\nu$ = 21.22 eV). Raman spectra for the 3F, 2F1N, and 1F2N electrolytes were acquired using a inVia Raman microscope (Renishaw) with a 532 nm excitation laser. Raman measurement for the 2F1N and 1F2D electrolytes were performed using a LabRAM HR Evolution (HORIBA) with a 785 nm excitation laser. The surface roughness of deposited lithium was analyzed using an NX-10 AFM instrument. Pre-SEI depth analysis was performed by XPS and TOF-SIMS. TOF-SIMS depth profiling was carried out using an ION-TOF instrument (Münster, Germany). 0.5 keV Cs^+ ion gun was used for sputtering the pre-SEI layer on the Cu current collector and 30 keV Bi_3^+ ion gun was used for analysis. XPS depth profiling was conducted with an ESCALAB 250 (Thermo Scientific).

5.5 | Computational Method

DFT calculations were performed using Vienna Ab-initio Simulation Package (VASP) software [42]. Core-valence interactions were described with the projector-augmented wave (PAW) method and a plane-wave basis set with a cutoff energy of 520 eV [43, 44]. The generalized gradient approximation (GGA) within Perdew-Burke-Ernzerhof (PBE) functional was used for the electronic exchange-correlation [45]. Electronic convergence criterion of 10^{-5} eV and the atomic force tolerance of 0.01 eV $Å^{-1}$ were used for all molecule calculations with the Γ-centered k-point-points mesh of 1 × 1 × 1. The GGA functional was used to obtain the initial solvation structures of various Li and additive complexes. To obtain accurate electronic structures of the molecular complex, the Hybrid functionals of Heyd-Scuseria-Ernzerhof (HSE06) was applied with the same electronic and atomic force tolerance [46, 47]. Calculations for the isolated molecules were conducted in a large cubic supercell with sufficient vacuum to minimize image interactions. LUMO, HOMO, and charge densities were computed with the VASPKIT software [48]. To visualize the charge densities of LUMO, HOMO, and molecule, the Vesta software was used [49].

MD simulations were conducted by LAMMPS software package [50]. The OPLS-AA force field [51] was applied to describe the TTE and TMGDME. For ionic species, the force field parameters for Li^+, FSI^-, NO3$^-$, and $DFOB^-$ ions were obtained from the literature: Dang [52], Gouveia et al. [53], Mendez-Morales et al. [54], and Jiang et al. [55], respectively. Atomic partial charges were derived using restrained electrostatic potential (RESP) [56, 57]. Atomic charges of Li^+, FSI^-, NO_3^-, and $DFOB^-$ were scaled by a factor of 0.8 to account for electronic polarization effect and attain the accurate density of electrolyte. The number of molecules in each electrolyte system was set according to experimental compositions (Table S2). Initial cubic boxes (75 × 75 × 75 $Å^3$) were constructed by PACKMOL package [58] and moltemplate [59]. The cutoff distances for Lennard-Jones (LJ) and Coulombic interactions were both set to 1.2 nm, and long-range electrostatic interactions were treated using the particle–particle particle-mesh (PPPM) method [60].

All simulations were performed with a time step of 1 fs following three-stage equilibration process. First, an initial minimization was performed in the isothermal-isobaric ensemble (NPT) at a temperature of 298.15 K and a pressure of 1 atm for 1 ns. Subsequently, a thermal annealing procedure was applied in the canonical ensemble (NVT) for a total of 1.5 ns: heating from 298.15 to 500 K (0.5 ns), maintaining at 500 K (0.5 ns), and cooling from 500 K to 298.15 K (0.5 ns). The production simulations were then carried out in the NVT ensemble at 298.15 K for 7.5 ns. RDF analysis was computed by averaging results obtained every 10 ps throughout the production phase.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: adma72305-sup-0001-SuppMat.docx.